Synthesizing Radiomics and Deep Learning for Oncological Prognosis: A Review of Methodological Evolution, Multimodal Integration, and Clinical Translation

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The integration of radiomics and deep learning (DL) has fundamentally transformed oncological prognosis, shifting the paradigm from subjective visual assessment to quantitative, data-driven precision medicine. This review synthesizes current literature on the evolution from handcrafted features to deep learning architectures, highlighting the emerging consensus that hybrid models combining both approaches often yield superior predictive performance by balancing interpretability with representational power [109] [60]. We examine the critical shift toward multimodal integration, where radiogenomic frameworks fuse imaging data with genomic and clinical variables to enable non-invasive "virtual biopsies," though contradictions persist regarding the predictability of specific molecular markers like MGMT methylation [100] [186]. Furthermore, we address the significant barriers to clinical adoption, including the "black-box" nature of deep models and data heterogeneity. Recent advancements in explainable AI, such as concept bottleneck models and neuro-symbolic architectures, are discussed as vital solutions for enhancing clinician trust and ensuring robustness across diverse imaging protocols [44] [192]. Ultimately, this review underscores that while technical accuracy has improved, future progress depends on rigorous standardization, reproducibility, and the seamless integration of these tools into clinical workflows to realize true precision oncology [272] [1].

Introduction to Radiomics and Deep Learning in Oncological Prognosis

The integration of radiomics and deep learning (DL) has fundamentally transformed oncological care, shifting the paradigm from subjective visual assessment to quantitative, data-driven precision medicine. This evolution enables the non-invasive extraction of high-dimensional features from medical images to predict diagnosis, prognosis, and treatment response across diverse malignancies. While traditional radiomics relies on handcrafted features describing tumor heterogeneity, DL automates feature extraction through convolutional neural networks (CNNs), often surpassing manual methods in complex tasks. Recent literature emphasizes a critical shift from single-modality analysis to multimodal frameworks that fuse imaging data with clinical, genomic, and metabolomic information to enhance predictive accuracy [18].

Traditional radiomics involves the extraction of predefined mathematical descriptors, such as texture, shape, and intensity statistics, from segmented regions of interest. These handcrafted radiomics (HCR) features offer high interpretability and have demonstrated efficacy in characterizing tumor phenotypes invisible to the human eye [5]. However, the reliance on manual segmentation introduces inter-observer variability, which can compromise the robustness of predictive models [7]. To address these limitations, deep learning radiomics (DLR) utilizes CNNs to automatically learn hierarchical feature representations directly from image data. Comparative studies indicate that DLR models often outperform conventional radiomics in prognostic tasks, such as predicting progression-free survival in nasopharyngeal carcinoma, by capturing more complex spatial and morphological patterns [27]. Furthermore, hybrid approaches that combine handcrafted and deep features have shown promise in reducing redundancy and enhancing predictive power, as evidenced by methods that minimize mutual information between feature sets to ensure complementarity [109]. Recent applications in neuro-oncology further validate that synergizing learned image representations with domain-targeted radiomic features provides a robust solution for automated treatment response assessment [60].

A significant trend in contemporary research is the move toward multimodal fusion, integrating imaging data with genomic, transcriptomic, and clinical variables to create comprehensive "virtual biopsies." This radiogenomic approach allows for the non-invasive prediction of molecular subtypes, such as EGFR and KRAS mutations in lung cancer, and MGMT promoter methylation status in glioblastoma, thereby guiding personalized treatment strategies without invasive procedures [155] [100]. By linking quantitative imaging features with genomic biomarkers, these frameworks can forecast disease progression and treatment resistance with high sensitivity [18]. For instance, models integrating histopathology whole slide images with genomic data have demonstrated superior performance in survival risk stratification across multiple cancer types compared to unimodal approaches [198].

Despite these advancements, the clinical adoption of DL models is frequently hindered by their "black-box" nature, which limits interpretability and clinician trust. To mitigate this, researchers are developing interpretable machine learning models that incorporate concept bottleneck architectures or fuse DL features with explainable radiomic signatures. Such approaches, like the ConRad model for lung cancer, combine concept bottleneck models with radiomics to provide transparent predictions that align with clinical reasoning [44]. Additionally, explainable AI techniques such as Grad-CAM and SHAP are increasingly employed to visualize model decisions and identify biologically relevant regions, ensuring that predictions are grounded in meaningful pathological or anatomical features [208] [155].

The challenge of data heterogeneity and limited annotated datasets remains a persistent obstacle, particularly in low-resource settings or for rare cancers. Strategies such as multi-task learning, transfer learning, and the use of lightweight architectures like MobileNetV2 are being explored to improve model generalization and reduce computational demands [72] [147]. Moreover, robust frameworks employing vector quantization and adaptive fusion mechanisms are being developed to maintain performance despite noise and protocol variations across different imaging centers [128]. As the field progresses, the synthesis of DL automation with the interpretability of traditional radiomics, alongside the integration of diverse data modalities, represents the forefront of developing reliable, clinically actionable prognostic tools [173].

The Evolution from Handcrafted Radiomics to Deep Learning

The trajectory of quantitative medical image analysis has undergone a paradigm shift, moving from a reliance on manually engineered descriptors to the automated extraction of hierarchical features via deep learning architectures. Traditional radiomics is predicated on the manual or semi-automated segmentation of regions of interest, followed by the computation of statistical, shape, and texture features which are subsequently fed into machine learning classifiers. While this approach has demonstrated robustness in specific clinical scenarios, it is inherently labor-intensive and susceptible to inter-observer variability during the segmentation phase [7]. Furthermore, the reproducibility of handcrafted features can be severely compromised by variations in imaging protocols, scanner manufacturers, and segmentation delineations. Studies have shown that even minor modifications to tumor volumes, such as smoothing or erosion, can drastically alter feature stability and predictive performance, with only a fraction of selected features remaining consistent across different delineation strategies [124]. Despite these limitations, handcrafted radiomics features remain valuable, particularly for tasks requiring high interpretability or when dealing with smaller datasets where deep learning models risk overfitting [5].

In contrast, deep radiomics utilizes Convolutional Neural Networks (CNNs) and other deep learning architectures to automatically learn feature representations directly from image data, thereby reducing the reliance on manual feature engineering and predefined mathematical formulas. This data-driven approach allows models to capture complex, non-linear patterns and hierarchical structures that may be invisible to human observers or traditional statistical methods [18]. Studies indicate that deep learning models often achieve superior performance compared to handcrafted approaches when large, annotated datasets are available. For instance, in the prediction of distant metastases in soft-tissue sarcomas, multi-modality deep learning architectures have outperformed state-of-the-art radiomic methods by effectively leveraging complementary information from PET and CT scans [249]. Similarly, in nasopharyngeal carcinoma, end-to-end deep learning-based radiomics models have demonstrated higher prognostic accuracy for progression-free survival than optimal conventional radiomics signatures [27].

However, the transition to deep learning is not without challenges, particularly regarding data scarcity and the opaque nature of these models. Deep learning algorithms are notoriously data-hungry, and their performance can degrade significantly in small cohorts or when faced with domain shifts across different imaging centers [72]. To mitigate these issues, recent research has increasingly focused on hybrid approaches that fuse handcrafted radiomics features with deep learning-derived features. This fusion strategy aims to combine the interpretability and robustness of engineered features with the representational power of deep networks. For example, in glioblastoma survival prediction, models integrating radiomics features, deep features, and patient-specific clinical data have shown improved accuracy over models using any single data source in isolation [41]. Similarly, in pancreatic cancer detection, methods that explicitly minimize the mutual information between handcrafted and deep learning radiomics ensure that the combined feature set is non-redundant, leading to enhanced diagnostic performance [109]. In the context of colorectal cancer, radiomics-based artificial intelligence models have been extensively reviewed for their utility in diagnosis, metastasis detection, prognosis, and treatment response, highlighting the growing consensus that integrating diverse feature sets enhances clinical decision support [1].

The debate between handcrafted and deep features also extends to the realm of interpretability and clinical trust. While deep learning models offer high accuracy, their lack of transparency hinders clinical adoption. Consequently, frameworks have been proposed to integrate expert-derived radiomics with concept bottleneck models, creating interpretable classifiers that maintain competitive performance against black-box CNNs [44]. Conversely, other studies suggest that deep learning features, when properly regularized or combined with attention mechanisms, can provide unique insights into tumor heterogeneity and treatment response that handcrafted features miss, such as in the differentiation of radiation necrosis from tumor progression [28]. Moreover, the evolution of these methods includes the development of automated workflows where deep learning is used not only for feature extraction but also for the initial segmentation of tumors, thereby addressing the bottleneck of manual contouring that plagues traditional radiomics [229]. Recent systematic reviews highlight that while deep radiomic features often achieve the highest mean accuracy in outcome prediction tasks, fusion models tend to attain the highest Area Under the Curve, suggesting that the synergy between manual and automated features is currently the most promising direction [173]. Additionally, advancements in semi-supervised learning and neural architecture search are further bridging the gap, allowing deep models to learn optimal feature representations even with limited labeled data, thus overcoming one of the primary hurdles of pure deep learning approaches [295]. These developments underscore a critical trend towards personalized radiation therapy strategies, where response-related features are unlocked through advanced modeling to guide adaptive treatment plans [261]. Ultimately, the field is moving towards a paradigm where the distinction between handcrafted and deep radiomics becomes increasingly blurred, with integrated frameworks leveraging the strengths of both to deliver robust, reproducible, and clinically actionable prognostic tools.

The Imperative for Multimodal Data Integration

The evolution of oncological prognosis has shifted decisively from unimodal assessments toward sophisticated multimodal strategies that synthesize imaging data with electronic health records, genomic markers, and metabolomic profiles. This paradigm shift addresses the inherent limitations of imaging-only models by incorporating critical biological context, such as immune cell status or specific genetic mutations, which are indispensable for the realization of personalized medicine. Research consistently demonstrates that combining computed tomography (CT) or positron emission tomography (PET) images with clinical variables significantly improves concordance indices in survival prediction across various malignancies compared to unimodal approaches. The integration of these diverse data streams allows for a more holistic characterization of tumor heterogeneity and patient-specific risk factors, thereby enhancing the robustness of prognostic algorithms.

In the realm of lung cancer, the fusion of radiological and genomic data has yielded substantial improvements in predicting recurrence and survival. Studies utilizing linear Cox proportional hazards models with elastic net regularization have shown that combining CT images with genomics can increase concordance-index values by up to 10% in non-small cell lung cancer (NSCLC) patients, although non-linear neural network methods have not always provided consistent gains, highlighting the need for larger datasets and adapted fusion techniques [29]. Furthermore, innovative integration methodologies synthesizing fused medical imaging with clinical health records and genomic data have leveraged advanced machine learning models like MedClip and BEiT to achieve classification accuracies exceeding 94%, setting a new standard in computational oncology [263]. Beyond simple fusion, recent advancements employ self-supervised modules and cross-patient modules to align embeddings across modalities, ensuring that semantic similarities between patients with similar disease characteristics are effectively captured to improve survival outcome predictions [296]. The utility of multimodal data extends to predicting specific molecular markers non-invasively; for instance, radiogenomics pipelines have successfully predicted EGFR and KRAS mutation statuses from CT scans, offering a viable alternative when tissue genotyping is invasive or delayed [155]. Additionally, genotype-guided radiomics methods have demonstrated the ability to improve prediction accuracy significantly by estimating gene expression from radiomic features, bridging the gap between low-cost imaging and high-cost genomic testing [103].

The imperative for multimodal integration is equally pronounced in head and neck cancer and other thoracic malignancies, where the combination of structural CT data and metabolic PET information provides complementary features critical for accurate prognosis. Deep learning-based approaches leveraging Convolutional Block Attention Modules to fuse CT and PET imaging modalities have demonstrated superior performance over conventional statistical models in predicting survival outcomes for head and neck cancer patients [15]. Similarly, end-to-end multi-modality deep learning-based radiomics models using pretreatment PET/CT images have outperformed single-modality models and conventional radiomics in predicting progression-free survival for advanced nasopharyngeal carcinoma, achieving higher area under the curve metrics in both internal and external cohorts [27]. In renal cell carcinoma, multimodal deep learning frameworks integrating 3D image feature extractors with systematically selected clinical variables have achieved concordance indices of 0.84, surpassing current literature based on CT scans or clinical factors alone [11]. These findings underscore that the synergistic use of anatomical and functional imaging, coupled with clinical metadata, creates a more discriminative feature space for risk stratification.

Addressing the complexity of metastatic and multifocal disease further necessitates advanced multimodal and multi-instance learning strategies. Traditional approaches often focus solely on the primary gross tumor volume, overlooking the prognostic implications of smaller lesions or lymph node metastases. Research indicates that employing multiple instance learning with injective pooling functions to analyze features from all tumors and metastases significantly improves outcome prediction compared to largest-lesion-based approaches [66]. Moreover, the integration of longitudinal data adds a temporal dimension to multimodal analysis, revealing patterns of disease progression and treatment response that are invisible in cross-sectional studies. Statistical modeling of longitudinal radiomic features derived from image registration-aided CT data has shown that incorporating additional imaging time points improves predictive performance, increasing mean concordance indices and revealing significant associations between feature evolution and survival outcomes [110].

The scope of multimodal integration is expanding to include pathological images, transcriptomic data, and even unconventional data sources such as pathology reports and facial photographs. Frameworks like PS3 utilize prototype-based representations to balance and fuse whole slide images, transcriptomics, and pathology reports, outperforming unimodal baselines by modeling intra-modal and cross-modal interactions effectively [PS3: A Multimodal Transformer Integrating Pathology Reports with Histology Images and Biological Pathways for Cancer Survival Prediction]. Hierarchical fusion frameworks that model the biological progression from genes to proteins to histology images from a systems biology perspective have also demonstrated superiority over flat fusion methods, capturing inherent biological hierarchies [182]. Even pan-cancer prognosis models are now integrating histopathology, genomics, and textual metadata regarding demographics and treatment protocols into unified networks, challenging the per-cancer-per-model paradigm and demonstrating strong generalization capabilities [185]. Additionally, novel approaches have explored the prognostic value of 2D facial photographs using StyleGAN embeddings, achieving meaningful concordance indices in pan-cancer analysis and suggesting that observable physical appearances contain latent health attributes predictive of survival [47]. Despite these advancements, challenges remain regarding data heterogeneity, the need for standardized reporting, and the clinical utility of these complex models, as systematic reviews indicate that many studies still suffer from methodological bias and limited external validation [Machine learning-based multimodal prognostic models integrating pathology images and high-throughput omic data for overall survival prediction in cancer: a systematic review]. Nevertheless, the trajectory of research clearly points toward increasingly complex, biologically informed, and clinically relevant multimodal integrations as the cornerstone of future precision oncology.

Methodological Frameworks for Feature Extraction and Segmentation

The accuracy and clinical utility of radiomic and deep learning models are intrinsically linked to the precision of tumor segmentation and the robustness of the subsequent feature extraction pipelines. Variability in segmentation, whether arising from inter-observer differences in manual contouring or inconsistencies in automatic algorithms, introduces significant noise that can compromise downstream predictive performance. Consequently, recent research has pivoted toward developing fully automatic segmentation networks, such as nnU-Net and various 3D Convolutional Neural Networks (CNNs), while rigorously evaluating the stability of features extracted from these automated masks against manual ground truths. The application of the nnU-Net for automatic segmentation of lung lesions on CT images has demonstrated that deep learning methods can mitigate the time-consuming nature and variability of manual segmentation, thereby enhancing the robustness of survival radiomic models [229]. Similarly, in the context of glioblastoma, fully automatic workflows utilizing 3D CNNs have been proposed to generate tumor contours from multimodal MR images, achieving high Dice coefficients and enabling more consistent survival predictions compared to handcrafted signatures derived from variable manual inputs [7].

Despite the promise of automation, the trade-off between segmentation accuracy and feature stability remains a critical area of investigation. Studies indicate that while the Dice Similarity Coefficient (DSC) is a standard metric for segmentation quality, it may not fully capture the subtle variations that impact radiomic feature reproducibility. Research suggests that radiomics features, particularly those related to shape and energy, exhibit greater sensitivity to segmentation changes than DSC, proposing that feature-based metrics like the Intraclass Correlation Coefficient (ICC) serve as superior measures for evaluating segmentation performance [171]. This sensitivity is further highlighted in breast cancer studies, where simulated variations in tumor delineation, such as erosion, smoothing, and dilation, resulted in significant shifts in selected radiomic features, with only a small fraction of features remaining robust across different volume of interest modifications [124]. These findings underscore the necessity of developing segmentation methods that not only maximize geometric overlap but also preserve the textural and statistical integrity required for reliable feature extraction.

To address the limitations of purely data-driven deep learning approaches, hybrid frameworks integrating handcrafted radiomics with deep learning features have emerged as a powerful methodological strategy. In liver tumor segmentation, a hybrid framework coupling an attention-enhanced cascaded U-Net with handcrafted radiomics and voxel-wise 3D CNN refinement has shown superior performance in handling low contrast and heterogeneous enhancement patterns [170]. This integration allows for the leveraging of expert-derived characterizations alongside learned representations. For instance, the ConRad model integrates concept bottleneck models to predict tumor biomarkers which are then combined with radiomic features, offering an interpretable alternative to black-box CNNs while maintaining high classification accuracy for lung nodule malignancy [44]. Furthermore, efforts to ensure that deep learning radiomics do not merely replicate hand-crafted radiomics have led to novel architectures, such as Variational Autoencoders that minimize mutual information between deep learning and hand-crafted features, ensuring the extraction of non-redundant, complementary information for early cancer detection [109].

The evolution of segmentation methodologies also extends to multi-task learning paradigms where segmentation and outcome prediction are performed jointly. The DeepMTS framework exemplifies this by using a hard-sharing segmentation backbone to guide feature extraction for survival prediction in nasopharyngeal carcinoma, effectively reducing background interference while capturing global tumor information [31]. This approach has been extended to head and neck cancer, where radiomics-enhanced deep multi-task frameworks extract handcrafted features from automatically segmented regions to improve prognostic stratification [3]. However, challenges persist regarding the generalizability of these models across different centers and imaging protocols. Robust frameworks like RobSurv utilize vector quantization to create noise-resistant representations from heterogeneous CT and PET images, demonstrating that resilient feature learning is essential for maintaining performance across diverse clinical settings [128]. Additionally, the development of segmentation-free approaches using multi-angle maximum intensity projections offers an alternative pathway that eliminates the dependency on subjective delineation entirely, though such methods must be carefully weighed against the detailed spatial insights provided by explicit segmentation [203].

The interplay between segmentation quality and predictive validity is further complicated by the specific requirements of different cancer types and imaging modalities. In brain tumor analysis, the heterogeneity of gliomas necessitates advanced architectures like Attention-Gated Recurrent Residual U-Nets to accurately delineate sub-regions, which are then used to extract geometric and textural features for survival prognosis [149]. Conversely, in scenarios where manual annotation is prohibitively expensive, semi-supervised learning strategies utilizing pseudo-labeling have proven effective in expanding training datasets and improving the stability of radiomic models for lung cancer prognosis [23]. In the realm of colorectal cancer, automated segmentation pipelines leveraging promptable foundation models have been developed to predict survival in liver metastasis, demonstrating the scalability of these methods to complex, multifocal diseases [176]. Furthermore, deep learning algorithms applied to histopathology images have enabled the automatic quantification of tumor-infiltrating lymphocytes, providing prognostic information that complements imaging-based radiomics [74]. The convergence of these methodologies points toward a future where segmentation is not merely a preprocessing step but an integral, optimized component of the predictive pipeline, designed specifically to maximize the fidelity of the extracted biological signatures.

Automated Segmentation Architectures and Performance

Deep learning architectures have fundamentally transformed the landscape of automatic tumor segmentation, establishing themselves as the standard for multi-center studies across various oncological domains. The transition from manual delineation to automated pipelines is driven by the critical need to reduce inter-observer variability and the substantial time burden associated with expert annotation, which often bottlenecks radiomic workflows. Among the myriad of architectures proposed, 3D U-Net variants and the nnU-Net framework have emerged as particularly robust solutions, offering a balance between architectural flexibility and automated preprocessing. In the context of head and neck cancer, the application of 3D nnU-Net has demonstrated significant efficacy, achieving mean Dice scores around 0.70 for the segmentation of primary tumors and lymph nodes [10]. This level of performance facilitates fully automatic pipelines that operate without the need for bounding box constraints, a critical advancement for clinical scalability and the handling of arbitrary fields of view [79]. Similarly, in glioblastoma research, 3D Convolutional Neural Networks (CNNs) have shown even higher precision, with average Dice coefficients reaching 0.85, generating contours that are sufficiently accurate to support subsequent survival analysis [7].

The architectural evolution in this field has moved beyond standard encoder-decoder structures to incorporate sophisticated mechanisms such as attention gates, dense connections, and multi-task learning frameworks. For instance, the integration of attention mechanisms into U-Net architectures, such as the Attention-Gated Recurrent Residual U-Net (R2U-Net), has been shown to enhance feature representation and segmentation accuracy by suppressing irrelevant background features and highlighting tumor boundaries. These models have achieved Dice Similarity Scores of 0.900 for whole tumor segmentation in brain tumor datasets, demonstrating the value of attention in handling complex morphologies [149]. Furthermore, the adoption of dense connectivity patterns within 3D fully convolutional neural networks allows for diversified feature learning and improved gradient flow, which is essential for handling the heterogeneous nature of gliomas and their sub-regions like necrosis and edema [221]. These architectural refinements address practical challenges such as class imbalance and the diffused boundaries of tumors, often employing hybrid loss functions that combine Dice loss with focal or cross-entropy losses to optimize convergence and boundary adherence [73].

Despite the dominance of U-Net variants, alternative approaches and hybrid frameworks continue to push the boundaries of performance. Multi-task deep learning models have gained traction for their ability to simultaneously perform segmentation and predict clinically relevant parameters, such as IDH mutation status or tumor grade, thereby leveraging shared representations to improve generalization across diverse patient cohorts [242]. In liver tumor segmentation, hybrid frameworks coupling attention-enhanced cascaded U-Nets with handcrafted radiomics and voxel-wise 3D CNN refinement have achieved Dice coefficients of 0.88, outperforming competing methods by effectively addressing low lesion-parenchyma contrast and blurred boundaries [170]. Additionally, the exploration of Vision Transformer (ViT)-based models offers a promising divergence from purely convolutional approaches, with some studies indicating that ViT-based frameworks can outperform ensemble CNN models in concurrent tumor segmentation and outcome prediction tasks [79].

However, the performance of these automated methods is not uniform across all tumor subregions or imaging modalities. While whole tumor segmentation often yields high Dice scores, delineating specific sub-regions such as the enhancing tumor or tumor core remains challenging, with performance metrics varying significantly depending on the complexity of the tumor morphology and the quality of the input data [214]. To mitigate these inconsistencies, techniques such as automatic hard mining have been introduced, where the training process dynamically focuses on difficult cases by increasing the Dice similarity coefficient threshold for sample selection as epochs progress [214]. Furthermore, the robustness of segmentation models across different centers and scanners is a critical concern. Studies utilizing leave-one-out-center cross-validation have reported average Dice scores of 0.82 for head and neck tumor segmentation, highlighting the capability of modern architectures to generalize, although performance can drop when applied to external test sets with distinct acquisition protocols [73].

The integration of segmentation outputs into broader prognostic workflows underscores the utility of these architectures. Accurate automated contours serve as the foundation for extracting radiomic features, which are then used to construct survival prediction models. Research indicates that signatures derived from deep learning-based auto-contours can successfully stratify patients into prognostically distinct groups, often outperforming signatures based on handcrafted features extracted from manual or less accurate segmentations [7]. Moreover, the development of segmentation-free approaches using multi-angle maximum intensity projections suggests a potential paradigm shift, although traditional segmentation-based pipelines remain the predominant method for leveraging spatial tumor characteristics in survival analysis [203]. The continued refinement of these architectures, including the incorporation of physics-embedded feature learning and adaptive multi-modality fusion, promises to further enhance the reliability and clinical applicability of automated segmentation systems [192]. In the realm of colorectal cancer, deep learning models analyzing H&E images have also been utilized to stratify patients into risk subgroups that predict survival and benefit from adjuvant chemotherapy, demonstrating the broader applicability of these automated feature extraction principles beyond just radiological segmentation [85].

Stability of Radiomic Features Across Segmentation Variabilities

The reliability of radiomic features extracted from automatically segmented regions remains a paramount concern in the development of robust predictive models for oncology. As the field transitions from manual delineation, which is considered the gold standard yet prone to inter- and intra-operator variability, toward fully automated deep learning pipelines, understanding the stability of features across different segmentation networks is critical. Research indicates that the choice of segmentation method significantly influences the extracted feature set, with texture features generally demonstrating higher stability compared to shape features or those derived from specific tumor sub-regions. Specifically, texture features extracted from the whole tumor region in T1 and T1-Gd MRI sequences exhibit superior consistency across different segmentation networks when contrasted with features from enhancing tumor cores or shape descriptors [6]. This discrepancy arises because shape features are intrinsically tied to the precise boundary definition, making them highly sensitive to the minor spatial variations inherent in automatic segmentation algorithms, whereas texture features capture internal heterogeneity that is less dependent on exact contour precision.

To quantify and mitigate this non-physiological variability, stability filtering techniques have become an essential component of the methodological framework. The Overall Concordance Correlation Coefficient (OCCC) and the Intraclass Correlation Coefficient (ICC) are frequently employed to identify robust features that maintain consistency despite segmentation perturbations. Studies have demonstrated that applying stability filters based on OCCC can minimize variability introduced by multi-regional segmentation maps obtained from fully automatic Convolutional Neural Networks (CNNs), thereby significantly improving the predictive performance of downstream tasks such as IDH mutation prediction and Overall Survival (OS) classification [6]. Furthermore, radiomics features have been proposed as a superior metric for evaluating segmentation performance itself, often exhibiting greater sensitivity to subtle segmentation differences than the traditional Dice Similarity Coefficient (DSC). For instance, while DSC values may remain high across different segmentations, the ICC of radiomic features, particularly shape and energy features, can reveal significant discrepancies, suggesting that radiomics offers a more comprehensive evaluation of segmentation quality [171].

The impact of segmentation variability extends beyond feature selection to the ultimate performance of predictive models. Investigations into breast cancer radiomics have shown that simulating variations in tumor delineation through mathematical operations such as erosion, smoothing, and dilation can drastically alter the set of selected features. In some instances, only a small fraction of features selected from manual volumes of interest remain consistent when different delineation data are used, highlighting the fragility of feature selection processes that do not account for segmentation uncertainty [124]. Interestingly, while segmentation accuracy is crucial, some studies suggest that predictive performance may not always correlate directly with segmentation precision. For example, in the prediction of Triple-Negative Breast Cancer subtypes, segmentation accuracy did not significantly impact predictive performance, and an overreliance on stability metrics like ICC for feature selection might inadvertently exclude valuable predictive features that are robust within the context of the machine learning model itself [269].

Contradictions and nuances exist regarding the optimal strategy for handling segmentation variability. While some approaches advocate for strict stability filtering to ensure reproducibility, others argue that the integration of deep learning-based signatures can inherently overcome some limitations of handcrafted features derived from auto-contours. Deep learning-based signatures generated from automatic segmentations have shown improved patient stratification and survival prediction capabilities compared to handcrafted signatures, suggesting that end-to-end learning may better accommodate segmentation noise [7]. Additionally, the concept of tensor radiomics has emerged as a novel paradigm to address variability by generating multiple flavours of features through segmentation perturbations and varying parameters, rather than relying on a single static extraction. This approach leverages the variability itself to construct more robust signatures, demonstrating improved reproducibility and prediction accuracy in glioblastoma and other cancers [116].

The interplay between segmentation methods and feature stability is further complicated by the specific tumor sub-regions analyzed. In glioblastoma, features extracted from the whole tumor region tend to be more stable than those from the enhancing tumor or necrotic core, which are often smaller and more irregularly shaped, making them susceptible to segmentation errors [6]. Consequently, methodological frameworks must carefully consider the trade-off between the biological relevance of specific sub-regions and the technical stability of features extracted from them. The use of automated segmentation tools like nnU-Net has facilitated large-scale studies, yet the variability introduced by these tools necessitates rigorous stability assessment before clinical deployment [10]. Ultimately, ensuring the stability of radiomic features across segmentation variabilities requires a multifaceted approach combining advanced stability metrics, robust feature selection strategies, and potentially new paradigms like tensor radiomics to harness rather than merely filter out variability.

Handcrafted versus Deep Feature Extraction Strategies

The methodological landscape of feature extraction in neuro-oncology is broadly categorized into two distinct paradigms: handcrafted radiomics and deep feature extraction. Handcrafted radiomics relies on the explicit computation of quantitative descriptors using predefined mathematical formulas to characterize tumor phenotype. These features typically encompass first-order statistics describing intensity distributions, shape descriptors quantifying morphology, and texture features capturing spatial relationships between voxels, such as those derived from Gray-Level Co-occurrence Matrices (GLCM) [5]. Tools like PyRadiomics have standardized the extraction of these features, enabling the calculation of hundreds of descriptors including Haralick texture features, which have demonstrated strong predictive power for brain tumor survival when paired with Random Forest classifiers [5]. The interpretability of handcrafted features remains a primary advantage, as each metric corresponds to a specific biological or physical property, such as tumor heterogeneity or sphericity, facilitating clinical translation and hypothesis generation [14]. Furthermore, studies indicate that specific subsets of handcrafted features, particularly shape and texture descriptors, can achieve state-of-the-art performance in survival prognosis, sometimes outperforming more complex deep learning architectures in data-limited scenarios [69].

Conversely, deep feature extraction leverages Convolutional Neural Networks (CNNs) to automatically learn high-dimensional representations directly from imaging data. In this approach, activation maps from intermediate layers of pre-trained or fine-tuned networks serve as compact descriptors of tissue heterogeneity, capturing complex patterns that may be invisible to human observers or traditional engineering [25]. Deep radiomic features (DRFs), often derived by aggregating CNN activation maps within tumor regions, have shown significant efficacy in correlating imaging phenotypes with underlying molecular biology. For instance, DRFs extracted from 3D-CNNs have successfully predicted immune cell marker status, including Macrophage M1, Neutrophils, and T Cells Follicular Helper, in glioma patients, offering a non-invasive window into the tumor microenvironment [25]. When combined with clinical variables, these deep features have demonstrated superior capability in stratifying patients into short-term and long-term survival groups compared to immune markers or clinical data alone [8]. The data-driven nature of deep features allows them to adapt to specific datasets, potentially uncovering non-linear relationships that handcrafted methods might miss [109].

Despite their respective strengths, both strategies exhibit limitations that have spurred the development of hybrid fusion approaches. Handcrafted features are constrained by their reliance on prior knowledge and may fail to capture higher-order semantic information, while deep features often suffer from a lack of interpretability and require large annotated datasets to avoid overfitting [72]. To mitigate these issues, recent frameworks propose fusing handcrafted radiomics with deep learning embeddings. For example, hybrid models concatenating CNN latent embeddings with radiomics features have shown improved validation performance and robustness in brain tumor classification tasks, particularly in low-resource settings where computational efficiency and interpretability are paramount [72]. Similarly, in the context of glioblastoma post-resection survival prediction, integrating handcrafted radiomics, deep features, and patient-specific clinical variables into a unified support vector machine framework yielded higher accuracy than any single modality used in isolation [41].

The debate between these strategies also extends to the redundancy of information they capture. Some research suggests that deep learning features may be highly redundant with handcrafted radiomics if not explicitly regularized, limiting the value of their combination [109]. To address this, novel methods employ variational autoencoders (VAEs) with mutual information minimization to ensure that deep features provide complementary information to handcrafted descriptors. Furthermore, advanced architectures like Neural Architecture Search (NAS) are being utilized to automatically derive optimal multi-modality fusion strategies, reducing the dependence on manual feature engineering and expert intuition [Multi-Modality Information Fusion for Radiomics-based Neural Architecture Search]. In longitudinal analyses, such as predicting treatment response in glioblastoma, hybrid frameworks that combine fine-tuned ResNet encoders with extensive sets of engineered radiomic and volumetric features have achieved robust performance in classifying response categories defined by RANO criteria [60]. Ultimately, the choice between handcrafted and deep strategies, or their integration, depends on the specific clinical task, data availability, and the need for model interpretability versus predictive power.

Harmonization Techniques for Multi-Center Data Heterogeneity

The integration of radiomic data from multiple institutions is a cornerstone for developing robust prognostic models in oncology, yet it introduces significant variability due to differences in scanner manufacturers, acquisition protocols, and reconstruction kernels. This heterogeneity can severely degrade model generalizability, necessitating the application of statistical harmonization techniques prior to model training. Among the various strategies employed, feature-domain harmonization methods, particularly ComBat, have become increasingly prevalent in multi-center studies to correct for batch effects while preserving biological signal. In the context of non-small cell lung cancer (NSCLC), research has demonstrated that harmonizing handcrafted radiomics and foundation model features using ComBat, often in conjunction with reconstruction kernel normalization (RKN), significantly improves survival prediction accuracy across diverse centers [146]. The efficacy of these methods relies on the assumption that batch effects are additive or multiplicative and can be estimated from the data distribution, allowing models to perform consistently on unseen test centers.

However, the application of harmonization is not without controversy, particularly regarding the interplay between harmonized radiomic features and conventional clinical variables. While some studies suggest that combining radiomic and clinical features yields additive prognostic benefits, others indicate that this combination does not always improve performance without rigorous feature selection and harmonization. For instance, in head and neck cancer (HNC) studies utilizing multi-center PET/CT data, the integration of conventional clinical features with radiomic signatures did not consistently enhance prognostic performance when compared to models using either data type in isolation, suggesting that careful feature selection is paramount to avoid redundancy or noise introduction [10]. Similarly, other investigations into HNC prognosis have found that while radiomics-clinical combined models can outperform clinical-only models in specific contexts, such as predicting recurrence-free survival in anal squamous cell carcinoma, the benefits are highly dependent on the specific feature set and the robustness of the harmonization pipeline applied [132]. These contrasting findings underscore that harmonization alone is insufficient; it must be paired with strategic feature engineering to unlock the full potential of multimodal data.

The challenge of data heterogeneity extends beyond scanner differences to include variations in image reconstruction and dose levels, which directly impact feature stability. Generative models, such as conditional generative adversarial networks (CGANs) and encoder-decoder networks, have been proposed as a pre-processing step to denoise low-dose CT scans, effectively transforming them into full-dose equivalents to improve the reproducibility of radiomic features [Generative Models Improve Radiomics Performance in Different Tasks and Different Datasets: An Experimental Study]. This approach complements statistical harmonization by addressing image-domain variability before feature extraction occurs. Furthermore, the concept of "tensor radiomics" has been introduced to systematically incorporate multiple "flavours" of features generated through varying discretization parameters, segmentation perturbations, and fusion methods. This paradigm aims to enhance model robustness by leveraging the diversity of feature representations rather than relying on a single set of parameters, thereby mitigating the sensitivity of models to specific acquisition variabilities [116].

Despite the promise of these techniques, contradictions exist regarding the optimal workflow for multi-center data integration. Some researchers argue that deep learning-based approaches, which learn features directly from raw images, may inherently be more robust to certain types of heterogeneity than handcrafted radiomics, provided sufficient training data is available. However, studies comparing deep learning features with handcrafted radiomics in multi-center settings often reveal that handcrafted features, when properly harmonized using methods like ComBat, can generalize better to external cohorts than deep features, which may overfit to site-specific artifacts [155]. Additionally, the impact of segmentation variability on radiomic stability cannot be overlooked; even with harmonized images, differences in tumor delineation across centers can introduce significant noise. Research indicates that while segmentation accuracy affects feature extraction, the predictive performance of models may remain stable if robust feature selection techniques are employed, although this remains a critical area of investigation for ensuring reproducibility [269].

The complexity of harmonization is further compounded in studies involving longitudinal data or multiple regions of interest (ROIs). In breast cancer research, longitudinal tracking of lesions across serial CT scans requires advanced registration algorithms to establish correspondence before harmonization and modeling can occur, as uncorrected spatial misalignment can mimic biological progression [110]. Moreover, when aggregating radiomic data from multiple lesions within a single patient, such as in regionally disseminated diseases, the strategy for combining these features must account for inter-lesion heterogeneity. Approaches that aggregate feature vectors from all available lesions have shown to improve predictive ability compared to analyzing the primary tumor alone, but these methods require consistent harmonization across all lesion sites to be effective [134]. Ultimately, while harmonization techniques like ComBat are indispensable for multi-center radiomics, their success is contingent upon a holistic framework that addresses image quality, segmentation consistency, and the strategic integration of multimodal data to ensure that prognostic models remain robust and clinically translatable. Future directions must also consider the role of explainable AI in validating that harmonization preserves biologically relevant signals, as seen in emerging frameworks for personalized radiation therapy response prediction [261].

Multimodal Integration Strategies for Enhanced Prognosis

The convergence of imaging data with non-imaging modalities represents the frontier of precision oncology, moving beyond the limitations of single-modality analysis to capture a comprehensive view of tumor biology. By fusing radiomic signatures with clinical demographics, genomic data, and metabolomic profiles, researchers aim to construct prognostic models that significantly outperform those derived from imaging alone. A foundational challenge in this domain is the robustness and reproducibility of radiomic features themselves. The paradigm of Tensor Radiomics addresses this by systematically incorporating multiple flavours of features, generated through varying discretization parameters, segmentation perturbations, and fusion levels, into high-dimensional tensors. This approach, utilizing architectures like the TR-Net, has demonstrated improved accuracy in overall survival prediction for head and neck and lung cancers compared to traditional single-flavour radiomics, suggesting that the systematic variation of extraction parameters can optimize signature construction [116]. Furthermore, the integration of imaging with genomic data, known as radiogenomics, allows for the non-invasive prediction of molecular subtypes and treatment responses. Frameworks like RADIOHEAD utilize voxel intensity probability density functions from multimodal MRI to model associations with pathway enrichment scores in lower-grade gliomas, revealing significant links between imaging heterogeneity and specific biological pathways such as synaptic transmission [238]. Similarly, genotype-guided radiomics methods have been proposed to estimate gene expression from CT images to predict recurrence in non-small cell lung cancer, achieving higher accuracy than methods using either modality in isolation [103]. In colorectal cancer, efforts to establish reproducible radiomic signatures have focused on analyzing features extracted from variable slice thickness contrast-enhanced CT scans, demonstrating that pooling features from multiple extraction settings while thresholding for reproducibility can yield prognostic models comparable to those using optimal single settings [80].

Beyond optimizing feature extraction and radiogenomic mapping, the integration of pathology reports and histology images with transcriptomic data offers another powerful avenue for survival prediction. The PS3 framework exemplifies this by employing a prototype-based multimodal transformer that balances representations across whole slide images, pathology reports, and biological pathways. By converting heterogeneous data sources into compact diagnostic, histological, and pathway prototypes, PS3 effectively models cross-modal interactions, outperforming unimodal and standard multimodal baselines across multiple cancer cohorts [PS3: A Multimodal Transformer Integrating Pathology Reports with Histology Images and Biological Pathways for Cancer Survival Prediction]. This addresses the critical issue of modality imbalance, where data-rich modalities like gigapixel images might otherwise dominate the learning process. In a similar vein, hierarchical fusion frameworks such as HFGPI model the biological progression from genes to proteins to histology images, employing graph-aware cross-attention to explicitly capture gene-protein regulatory relationships and protein-morphology associations, thereby enhancing survival prediction accuracy [182].

Architectural strategies for fusing these diverse data streams are as critical as the data sources themselves. Deep learning architectures have evolved to handle complex multimodal inputs through sophisticated attention mechanisms and neural architecture search. For instance, the MM-NAS method automatically derives optimal convolutional neural network architectures for fusing PET and CT images, removing the reliance on manual design and improving the prediction of distant metastases in soft-tissue sarcomas [Multi-Modality Information Fusion for Radiomics-based Neural Architecture Search]. In the context of head and neck cancer, models integrating CT and PET imaging with clinical data via Convolutional Block Attention Modules and multi-modal fusion layers have shown superior performance over traditional Cox proportional hazards models [15]. Furthermore, the inclusion of longitudinal data adds a temporal dimension to multimodal integration. Studies on metastatic breast cancer have utilized registration-aided automated correspondence algorithms to track lesions across serial CT scans, demonstrating that joint modeling of longitudinal radiomic features and hazard rates significantly improves progression-free survival prediction compared to baseline-only models [123]. To address the vulnerability of these complex models to noise, frameworks like RobSurv leverage vector quantization to create noise-resistant discrete codebooks, maintaining robust performance under severe noise conditions where baseline methods degrade significantly [128]. Additionally, generative models have been employed to denoise low-dose CT scans, transforming them into full-dose equivalents to improve the reliability of extracted radiomic features for survival prediction and diagnosis [Generative Models Improve Radiomics Performance in Different Tasks and Different Datasets: An Experimental Study].

Despite these advancements, contradictions and limitations persist regarding the added value of complex radiomic features over simple clinical variables. A large multi-center study on glioblastoma found that while combined models of conventional radiomics and clinical data performed well, standard radiomic approaches offered minimal added prognostic value beyond basic demographic information like age and gender for short-term survival prediction [32]. This contrasts sharply with findings in other domains, such as NSCLC, where harmonized multi-region radiomics combined with foundation model features significantly improved survival prediction over clinical staging alone [146]. The synthesis of these varied approaches highlights that while multimodal integration holds immense promise, the choice of fusion architecture, the handling of data heterogeneity, and the specific clinical context dictate the extent of prognostic improvement. Future directions point towards more interpretable and biologically informed models, such as those incorporating immune cell markers into deep radiomic signatures to predict immunotherapy response in glioma patients [8], and the development of standardized, reproducible pipelines like PySERA to ensure the scalability of these complex integrations [130].

Fusion of Imaging and Clinical Demographics

The integration of clinical demographics with advanced imaging features represents a pivotal strategy in modern oncological prognosis, moving beyond unimodal limitations to capture the multifaceted nature of tumor biology and patient health. This multimodal approach leverages the complementary strengths of quantitative imaging data-such as texture, shape, and metabolic activity-and established clinical variables like age, gender, smoking status, and TNM stage. In the context of renal cell carcinoma, research has demonstrated that combining 3D convolutional neural network (CNN) image features with systematically selected clinical variables yields superior predictive performance compared to models relying solely on CT scans. Specifically, a multimodal deep learning framework utilizing a 3D CNN for feature extraction and random forest importance scores for clinical variable selection achieved a concordance index of 0.84, significantly outperforming existing literature and highlighting the critical value of demographic and historical patient data in survival analysis [11]. Similarly, in non-small cell lung cancer (NSCLC), the DeepMMSA framework exemplifies this synergy by leveraging CT images alongside clinical data to improve survival prediction accuracy. This method increased the percentage of concordant pairs by 4% compared to traditional statistical methods, validating the hypothesis that underlying relationships exist between prognostic information embedded in radiomic images and clinical histories [2].

The efficacy of fusing imaging with clinical data extends across various cancer types and imaging modalities, often addressing the limitations of single-modality approaches. For advanced nasopharyngeal carcinoma, multi-modality deep learning-based radiomics models integrating pretreatment PET/CT images with high-level clinical features, such as TNM stage, have shown enhanced prognostic performance over conventional radiomics. These models achieved higher Area Under the Curve (AUC) values in both internal and external cohorts, demonstrating that clinical staging provides essential context that refines the interpretation of deep imaging features [27]. Crucially, while clinical signatures alone failed to distinguish risk groups in external validation sets for this disease, the combination of deep learning radiomics signatures with clinical data maintained significant prognostic value, underscoring the necessity of multimodal integration to ensure model generalizability [27]. Furthermore, the integration of clinical variables into deep learning architectures allows for more robust risk stratification. In head and neck cancer, models that incorporate clinical data extracted from electronic medical records alongside pre-treatment radiological images have outperformed those relying exclusively on engineered radiomics or deep learning features alone. The winning approaches in institutional challenges often utilize non-linear, multi-task learning on combined clinical and imaging data, suggesting that simple, informative prognostic factors act as powerful regularizers or enhancers when fused with complex image-derived representations [26].

Despite the clear advantages, the methodology for integrating these heterogeneous data sources varies, with researchers exploring different fusion strategies to optimize performance. Some studies employ early fusion, where raw or low-level features are concatenated, while others utilize late fusion or decision-level integration. For instance, in pancreatic ductal adenocarcinoma, a deep learning fusion model integrating radiology reports (textual clinical data) and CT imaging achieved a concordance index of 0.6750 for 5-year survival risk estimation, significantly separating high and low-risk groups. This indicates that even unstructured clinical text can be effectively fused with voxel-level imaging data to enhance prognosis [118]. Conversely, other frameworks propose more sophisticated architectures, such as the Lite-ProSENet for NSCLC, which uses a two-tower architecture to separately model textual clinical data via a light transformer and visual CT scans via a 3D ResNet, before fusing them for survival prediction. This approach mimics human clinical decision-making and achieved a state-of-the-art concordance of 89.3%, surpassing methods that simply concatenate features [76]. Additionally, adaptive mechanisms are emerging, such as the AdaMSS framework, which optimizes fusion strategies dynamically during training based on the specific patient context and data availability, moving beyond static empirically-designed fusion networks [46].

However, the fusion of imaging and clinical demographics is not without challenges, particularly regarding data heterogeneity, interpretability, and the potential for overfitting in high-dimensional spaces. While deep learning models excel at extracting complex patterns, they often function as "black boxes," limiting clinical trust. To address this, some researchers advocate for interpretable machine learning models that integrate expert-derived radiomics and predicted biomarkers with clinical data. The ConRad model, for example, combines concept bottleneck models with radiomic features to classify lung nodules, offering transparency that pure deep learning approaches lack while maintaining competitive accuracy [44]. Additionally, the robustness of fused models across different institutions remains a concern. Studies have shown that handcrafted radiomic features combined with clinical data can sometimes generalize better than deep learning features when applied to external datasets, suggesting that the choice of fusion strategy must account for the specific constraints of the clinical environment and data availability [155]. Moreover, the inclusion of clinical variables helps anchor the high-dimensional feature space of medical images, reducing the curse of dimensionality and guiding the model toward biologically plausible associations. Ultimately, the consensus across diverse oncological domains is that the fusion of imaging and clinical demographics is not merely an additive process but a synergistic one, essential for developing precise, reliable, and clinically actionable prognostic tools.

Radiogenomics: Linking Imaging Phenotypes to Genomic Markers

Radiogenomics has emerged as a transformative discipline within neuro-oncology, striving to non-invasively infer genetic mutations and molecular subtypes directly from standard medical imaging. This approach effectively bridges the critical gap between radiological phenotypes and pathological genotypes, offering a scalable alternative to invasive biopsies which are often limited by sampling bias and procedural risks. In the context of brain tumors, particularly glioblastoma (GBM), the ability to predict key biomarkers such as Isocitrate Dehydrogenase (IDH) mutation status and 1p/19q co-deletion from multi-parametric MRI scans is fundamental for accurate prognosis and therapeutic decision-making. Recent systematic reviews and meta-analyses have validated the efficacy of deep learning models in this domain, reporting pooled area under the curve (AUC) estimates of 0.89 for IDH prediction and 0.90 for 1p/19q codeletion, demonstrating high diagnostic performance across diverse patient cohorts [Diagnostic Performance of Deep Learning for Predicting Gliomas' IDH and 1p/19q Status in MRI: A Systematic Review and Meta-Analysis]. Furthermore, advanced multi-task deep learning frameworks have been developed to simultaneously segment tumors and predict these molecular statuses, achieving robust generalization across multiple institutes by leveraging full 3D structural information from pre-operative scans [242].

Beyond the binary classification of mutation status, the integration of radiomic features with gene expression data has significantly enhanced overall survival (OS) estimation. Traditional models relying solely on radiomic features often fail to capture the underlying biological drivers of tumor aggressiveness. However, studies utilizing The Cancer Genome Atlas (TCGA) dataset have demonstrated that fusing gene expression levels with radiomic features-encompassing shape, geometry, and texture-substantially improves OS prediction in GBM patients compared to unimodal approaches [21]. This fusion strategy is complemented by novel feature extraction methods such as spherical radiomics, which captures the radial growth patterns and intratumoral heterogeneity of GBM more effectively than conventional Cartesian grids. By extracting features from concentric shells, this approach has shown superior correlation with molecular markers like MGMT promoter methylation, EGFR, and PTEN mutation status, achieving AUCs of 0.85, 0.80, and 0.80 respectively, alongside improved survival prediction capabilities [106].

A persistent challenge in radiogenomic modeling is the incompleteness of multimodal imaging data in clinical settings, which can severely hinder model performance. To mitigate this, generative adversarial networks (GANs) have been employed to synthesize missing MRI modalities, thereby enabling comprehensive radiomic analysis even when scans are incomplete. Fully convolutional networks within conditional GAN frameworks have successfully synthesized missing sequences, leading to improved tumor segmentation and subsequent radiogenomic survival predictions [21]. While synthetic data generation offers a promising solution for data augmentation and privacy preservation, its utility varies depending on the complexity of the anatomical structures, with synthetic MRIs showing higher fidelity and segmentation utility compared to synthetic CTs in certain contexts [178].

Despite these advancements, the field faces significant contradictions regarding the predictability of certain biomarkers, most notably MGMT promoter methylation. While some studies report promising results using hybrid frameworks that combine radiomics, deep learning, and explainable artificial intelligence (XAI) to predict MGMT status with high interpretability [100], rigorous benchmarks suggest that current models may be fundamentally unlearnable due to the lack of visible cues in MRI scans correlating with MGMT status. Investigations using extensive datasets like BraTS 2021 have concluded that there appears to be no discernible connection between MRI scans and the state of the MGMT promoter, suggesting that deep learning models might be capturing dataset biases rather than true biological signals [186]. This discrepancy underscores the necessity for large-scale, multi-center validation and the development of stability-aware modeling techniques to ensure that radiogenomic signatures reflect genuine biological phenomena [6].

To address the limitations of whole-tumor analysis and enhance clinical utility, recent efforts have focused on creating "virtual biopsy" maps that provide spatially resolved predictions of mutational status. These spatial-and-context aware (SpACe) maps combine context-aware features from co-localized biopsy sites with spatial priors derived from population atlases to generate voxel-wise probability maps of gene mutations. Such approaches have demonstrated higher accuracy in identifying EGFR amplification and MGMT methylation status compared to traditional whole-tumor models, offering potential guidance for surgical navigation and targeted sampling [Spatial-And-Context aware (SpACe) "virtual biopsy" radiogenomic maps to target tumor mutational status on structural MRI]. Additionally, the incorporation of semi-supervised learning strategies has improved the robustness of IDH mutation prediction, particularly in scenarios with limited labeled data, by leveraging unlabeled datasets to refine feature selection and model stability [295]. Collectively, these advancements highlight the potential of radiogenomics to serve as a non-invasive, comprehensive tool for precision oncology, provided that challenges related to data heterogeneity, model interpretability, and biological validity are systematically addressed.

Integration of Metabolomic and Immune Microenvironment Data

Moving beyond the static information provided by genomic sequencing, the frontier of oncological imaging is increasingly defined by the integration of dynamic metabolomic profiles and the spatial architecture of the immune microenvironment. This multimodal paradigm seeks to capture the functional physiology of tumors, leveraging the rich chemical data from tissue metabolomics and the contextual distribution of immune cells to enhance non-invasive diagnostic and prognostic capabilities. A seminal advancement in this domain is the development of the Tissue-Metabolomic-Radiomic-CT (TMR-CT) framework, which successfully embeds tissue metabolome information directly into Computed Tomography (CT) image representations. By training deep learning models on paired CT images and metabolite intensities from both tumor and normal tissues, this approach generates image embeddings capable of inferring metabolite-derived representations from CT scans alone. In clinical applications for non-small cell lung cancer (NSCLC), TMR-CT has achieved state-of-the-art performance in histology classification and prognosis prediction, surpassing conventional radiomics and clinical features without the need for invasive biopsies. This methodology effectively bridges the gap between hard-to-obtain metabolic profiles and routine lesion-derived image data, suggesting that imaging can serve as a robust surrogate for complex biological assays [12].

Parallel to metabolomic integration, significant progress has been made in correlating deep radiomic features with the immune microenvironment, particularly within the context of gliomas. Research demonstrates that deep radiomic features (DRFs), computed from convolutional neural networks applied to multi-sequence MRI, can capture tumor characteristics associated with specific immune cell markers such as Macrophage M1, Neutrophils, and T Cells Follicular Helper. These features provide a compact yet powerful representation of regional texture that encodes tissue heterogeneity, showing high correlation with immune marker status. When combined with clinical variables, these DRFs enable the discrimination between short and long survival outcomes with high statistical significance, indicating their utility as non-invasive biomarkers for predicting immunotherapy response [8]. Further investigations utilizing Gaussian mixture models to analyze the distribution of learned 3D CNN features have reinforced these findings, demonstrating that DRFs can predict immune marker status with area under the curve values exceeding 75% for various cell types. The combination of these features with immune markers and clinical variables yields highly significant stratification of patient survival groups, outperforming models based solely on clinical variables or immune markers [25].

The broader landscape of radiogenomics supports the feasibility of using imaging signatures as proxies for the tumor-host immune apparatus. Systematic reviews indicate that imaging biomarkers extracted from MRI volumes, such as apparent diffusion coefficient values and image-derived features, correlate with genomic markers from tumor or immune cells and patient survival in glioblastoma. While these radiogenomic immune biomarkers hold the potential to stratify patients for immunotherapy clinical trials and facilitate personalized neo-adjuvant treatments, current literature notes a lack of prospective validation and concerns regarding study bias and generalizability [248]. Despite these challenges, the fusion of imaging with multi-omics data continues to evolve, with deep learning frameworks increasingly capable of linking quantitative imaging features with transcriptomics and epigenetic biomarkers to predict tumor genotype and immune response [18].

The synergy between different data modalities is further exemplified by studies integrating radiomics with other biological layers to enhance interpretability and robustness. For instance, the integration of radiomics and tumor biomarkers in interpretable machine learning models, such as ConRad, combines concept bottleneck models with expert-derived radiomic features to enhance transparency while maintaining high performance in malignancy classification [44]. Additionally, the concept of "tensor radiomics" proposes a systematic incorporation of multi-flavored radiomic features generated through varying parameters, which has shown improved reproducibility and performance in tasks involving glioblastoma and lung cancer, suggesting that robust feature extraction is critical for reliable biological inference [116]. The application of these principles extends to predicting treatment response, where hybrid models combining deep learning features with extensive radiomic and engineered features have achieved robust performance in classifying tumor response according to standardized criteria like RANO in glioblastoma [60].

Furthermore, the exploration of vascular morphology as a biomarker underscores the depth of information extractable from imaging. Quantitative tumor-associated vasculature (QuanTAV) features, which measure vessel twistedness and organization, have been shown to predict response to chemotherapy and survival across multiple cancer types, independent of clinicopathological risk factors [93]. This aligns with findings that deep radiomic signatures can predict survival in advanced nasopharyngeal carcinoma, outperforming conventional radiomics and clinical signatures alone [27]. The collective evidence from these diverse studies confirms that integrating metabolomic, immune, and vascular data with advanced imaging analytics provides a comprehensive, non-invasive window into tumor biology, offering profound implications for precision oncology and patient management.

Advanced Neural Architectures for Multimodal Fusion

The integration of heterogeneous data sources in oncological prognosis necessitates sophisticated neural architectures capable of dynamically weighing the importance of distinct modalities. Traditional statistical models, such as the Cox proportional hazards model, often fail to capture the complex, non-linear interactions inherent in multimodal datasets comprising imaging, genomic, and clinical variables. To address these limitations, recent advancements have focused on attention-based mechanisms and multi-stream networks that facilitate adaptive feature fusion. A prominent example of this evolution is the application of Convolutional Block Attention Modules (CBAM) within deep learning frameworks designed for head and neck cancer survival prediction. By employing CBAM, models can effectively fuse features from Positron Emission Tomography (PET) and Computed Tomography (CT) scans before integrating them with clinical data, thereby overcoming the linearity constraints of traditional approaches and significantly improving predictive accuracy [15]. This architecture allows the network to selectively emphasize salient features from different modalities, ensuring that metabolic information from PET and structural details from CT are optimally combined to inform the final survival risk estimation. Similarly, ensemble approaches combining deep multi-task logistic regression with CNNs have demonstrated superior performance in prognosis tasks by fusing imaging and electronic health records, achieving high concordance indices in challenge settings [24].

Beyond standard convolutional attention, the field has increasingly adopted Transformer-based architectures to model long-range dependencies and complex feature interactions across modalities. Vision Transformers (ViTs) have been adapted for survival analysis to recover outcome-relevant regions through token-level attention dynamics, offering a level of interpretability that pure convolutional networks often lack. In this context, radiomics-guided hybrid models integrate pixel-based embeddings with interpretable radiomic features via multimodal Cox frameworks, enabling the model to leverage both low-level visual patterns and high-level quantitative descriptors [194]. Similarly, cross-attention mechanisms have been utilized to fuse histopathological image features with genetic and clinical data in breast cancer prognosis. These dual cross-attention modules allow for the intricate modeling of inter-modal relationships, where image-derived features are aligned with genomic signatures to generate a holistic patient profile, demonstrating superior performance over unimodal baselines [165]. The efficacy of such attention-driven fusion is further evidenced in lung cancer studies, where multi-head self-attention mechanisms within textual towers process clinical data while parallel visual towers extract compact features from 3D CT scans, simulating the integrative decision-making process of human clinicians [76]. Furthermore, hierarchical fusion frameworks are emerging to model biological progressions from genes to proteins to histology, utilizing graph-aware cross-attention to capture higher-order relationships [182].

A critical challenge in multimodal fusion lies in the heterogeneity of data scales and the varying relevance of spatial regions across different imaging modalities. To mitigate this, adaptive fusion strategies have been developed to self-optimize the combination of modalities during training. For instance, the Adaptive Multi-modality Segmentation-to-Survival (AdaMSS) model employs a data-driven strategy to fuse PET and CT information, allowing the network to adaptively shift its focus from tumor-centric regions to broader prognostic areas, such as lymph node metastasis, across different training stages [46]. This adaptability is crucial because empirically designed fusion strategies, such as simple early or late fusion, often fail to account for scenario-specific variations in data quality and relevance. Furthermore, multi-scale cross-attention frameworks have been introduced to handle the vast granularity of Whole Slide Images (WSIs), where features extracted at different magnification levels interact to capture both cellular details and global tissue architecture [247]. By modeling interactions between coarse and fine scales, these architectures ensure that complementary information is preserved and effectively utilized for survival prediction. Graph-based approaches also contribute by modeling spatial relationships between patches or lymph nodes, providing context-aware representations that outperform standard multiple instance learning [253] [153].

The integration of radiomics with deep learning features represents another significant avenue for enhancing multimodal fusion. While deep learning models excel at automatic feature extraction, they often lack the interpretability of handcrafted radiomic features. Hybrid approaches address this by fusing deep features with radiomic descriptors through variational autoencoders or dedicated fusion layers, as seen in frameworks designed for ER+ breast cancer risk stratification [284]. This synergy allows the model to benefit from the robustness of learned representations while retaining the clinical interpretability of radiomics. Moreover, semi-supervised learning strategies have been employed to leverage unlabeled data in multimodal settings, where pseudo-labeling techniques enhance the training of models that fuse clinical, PET, and CT data, significantly reducing prediction errors compared to supervised-only approaches [42] [52].

Despite these advancements, contradictions and gaps remain regarding the optimal fusion topology and the handling of missing modalities. Some studies suggest that non-linear neural network fusion methods do not consistently outperform regularized linear models when datasets are small or highly heterogeneous, indicating a need for larger, more diverse multimodal cohorts [29]. Additionally, the reliance on complete multimodal data limits clinical applicability, prompting the development of robust frameworks like RobSurv, which utilizes vector quantization to maintain performance under noise and protocol variations across imaging centers [128]. Future research must continue to refine these adaptive architectures to ensure they can generalize across diverse cancer types and imaging protocols while maintaining the interpretability required for clinical adoption.

Conclusion and Summary of Key Findings

The synthesis of radiomics and deep learning has undeniably advanced the field of oncological prognosis, offering non-invasive, accurate, and personalized insights into tumor behavior. From the automation of segmentation to the fusion of multimodal data, these technologies have demonstrated superior performance in predicting survival and treatment response across colorectal, lung, head and neck, and brain cancers. However, the journey to widespread clinical adoption requires continued focus on standardization, interpretability, and robust validation. As the field evolves towards radiophenomics and federated learning, the potential to transform cancer care into a truly precision-based discipline becomes increasingly tangible.

The integration of handcrafted radiomic features with deep learning-derived representations has emerged as a dominant paradigm, often outperforming unimodal approaches. Studies indicate that while deep learning models excel at capturing abstract, hierarchical patterns, handcrafted features provide interpretability and stability, particularly in scenarios with limited data. For instance, research on brain tumor survival prediction highlights that radiomics-based approaches using engineered features can achieve state-of-the-art accuracy, sometimes surpassing pure deep learning methods which may suffer from instability in survival tasks [5]. Similarly, in head and neck cancer, frameworks that combine deep multi-task learning for segmentation with automatic radiomics extraction have shown significant improvements in outcome prediction, effectively leveraging both global tumor information and localized texture features [3]. This synergy is further validated in glioblastoma prognosis, where hybrid models fusing deep features, radiomics, and clinical data yield higher accuracy than single-modality inputs [41]. Furthermore, strategies to ensure non-redundancy between hand-crafted and deep learning radiomics have been shown to improve early detection capabilities, as demonstrated in pancreatic cancer studies where mutual information minimization enhanced feature complementarity [109].

A critical advancement in this domain is the shift towards multi-modal and pan-cancer frameworks that transcend single-disease limitations. Traditional models often operate under a "per-cancer-per-model" paradigm, restricting generalizability. Recent innovations, such as unified multi-modal networks, successfully integrate histopathology, genomic profiles, and clinical text data to perform joint prognostic analysis across diverse cancer types, demonstrating that collective learning enhances model robustness [185]. Furthermore, the incorporation of biological knowledge into machine learning pipelines, termed knowledge-informed machine learning, addresses challenges related to data heterogeneity and model interpretability, ensuring that predictions align with established biomedical principles [35]. The fusion of imaging with genomics, or radiogenomics, has also proven vital; for example, models predicting EGFR mutation status in lung cancer or MGMT methylation in glioblastoma utilize deep radiogenomic pipelines to non-invasively infer molecular biomarkers, potentially reducing the need for invasive biopsies [77] [100]. However, contradictions exist; while some pipelines claim success, other rigorous analyses using extensive datasets like BraTS 2021 conclude there may be no discernible connection between MRI scans and MGMT promoter status, highlighting the need for cautious interpretation [186]. Additionally, robust deep-learning frameworks leveraging vector quantization have been introduced to handle noise and protocol variations across imaging centers, significantly improving survival prediction consistency in heterogeneous datasets [128].

Despite these successes, significant challenges remain regarding standardization, reproducibility, and the "black-box" nature of complex algorithms. Systematic reviews of PET and SPECT imaging studies reveal that while deep radiomic features often achieve high accuracy, issues such as class imbalance, missing data, and low population diversity persist, with fewer than half of studies adhering to international biomarker standardization initiatives [173]. The variability in tumor delineation also poses a risk; variations in volume of interest definition can significantly alter feature extraction and prediction performance, underscoring the need for automated, robust segmentation tools [124]. To mitigate these issues, researchers are developing interpretable models like ConRad, which integrate concept bottleneck models to predict biomarkers alongside radiomic features, thereby enhancing transparency without sacrificing accuracy [44]. Additionally, semi-supervised learning strategies are gaining traction for their ability to leverage unlabeled data, improving model stability and performance in data-scarce environments [23] [52].

Future directions emphasize the development of adaptive, physics-embedded, and clinician-in-the-loop systems. Approaches that embed physical laws of tumor growth directly into neural network architectures, such as PhysNet, offer a pathway to more trustworthy and biologically consistent predictions [192]. Moreover, the transition from static analysis to longitudinal monitoring using delta-radiomics allows for the dynamic assessment of treatment response, as seen in studies differentiating radiation necrosis from tumor progression or predicting lesion shrinkage during therapy [28] [294]. The ultimate goal is to create deployable clinical tools that not only predict outcomes but also guide personalized treatment strategies, such as identifying patients who will benefit from adjuvant chemotherapy in colorectal cancer or optimizing radiation therapy plans based on spatially informative features [1] [261]. Specific applications in colorectal liver metastases have also shown that automated frameworks integrating segmentation and radiomics can deliver accurate, annotation-efficient outcome predictions [98] [176]. As the field matures, the focus must remain on rigorous external validation, diverse dataset inclusion, and the seamless integration of these advanced computational tools into clinical workflows to realize the full promise of precision oncology.